

Could not connect to the reCAPTCHA service. Please check your internet connection and reload to get a reCAPTCHA challenge.

```
1 from google.colab import drive
2 drive.mount("/content/drive")
3
```

Mounted at /content/drive

```
1 !mkdir -p /content/mudcadx_yolo/images/train
2 !mkdir -p /content/mudcadx_yolo/images/val
3 !mkdir -p /content/mudcadx_yolo/labels/train
4 !mkdir -p /content/mudcadx_yolo/labels/val
5
```

```
1 import os, glob, cv2
2 import numpy as np
3 import random
4
5 mud_base = "/content/drive/MyDrive/major_project/dataset" # your MUDCAD-X root
6 seasons = ["01_Spring", "02_Summer", "03_Autumn"]
7
8 out_base = "/content/mudcadx_yolo" # output in /content (FAST)
9
10 train_img = os.path.join(out_base, "images/train")
11 val_img = os.path.join(out_base, "images/val")
12 train_lbl = os.path.join(out_base, "labels/train")
13 val_lbl = os.path.join(out_base, "labels/val")
14
15 os.makedirs(train_img, exist_ok=True)
16 os.makedirs(val_img, exist_ok=True)
17 os.makedirs(train_lbl, exist_ok=True)
18 os.makedirs(val_lbl, exist_ok=True)
19
20 CAMO_CLASS_ID = 0
21 VAL_RATIO = 0.2
22 random.seed(42)
```

```

23
24 def mask_to_yolo_bbox(mask):
25     h, w = mask.shape
26     _, bin_mask = cv2.threshold(mask, 127, 255, cv2.THRESH_BINARY)
27     contours, _ = cv2.findContours(bin_mask, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)
28     if len(contours) == 0:
29         return None
30     x, y, bw, bh = cv2.boundingRect(max(contours, key=cv2.contourArea))
31     return (x + bw/2)/w, (y + bh/2)/h, bw/w, bh/h
32
33 def read_gray(p):
34     return cv2.imread(p, cv2.IMREAD_GRAYSCALE)
35
36 total = 0
37
38 for season in seasons:
39     season_path = os.path.join(mud_base, season)
40     label_paths = glob.glob(season_path + "/*/label.png", recursive=True)
41     print(f"{season} labels found: {len(label_paths)}")
42
43     for label_path in label_paths:
44         folder = os.path.dirname(label_path)
45
46         mask = read_gray(label_path)
47         if mask is None:
48             continue
49
50         bbox = mask_to_yolo_bbox(mask)
51         if bbox is None:
52             continue
53
54         # split train/val
55         if random.random() < VAL_RATIO:
56             save_img = val_img
57             save_lbl = val_lbl
58         else:
59             save_img = train_img
60             save_lbl = train_lbl

```

```

60         save_lbl = train_lbl
61
62     r_path = os.path.join(folder, "red.png")
63     g_path = os.path.join(folder, "green.png")
64     b_path = os.path.join(folder, "blue.png")
65     nir_path = os.path.join(folder, "nir.png")
66     eir_path = os.path.join(folder, "eir.png")
67
68     rel = os.path.relpath(folder, mud_base).replace("/", "_")
69     base_name = "mud_" + rel
70
71     def save_sample(img3, suffix):
72         global total
73         img_name = base_name + suffix + ".png"
74         cv2.imwrite(os.path.join(save_img, img_name), img3)
75
76         with open(os.path.join(save_lbl, img_name.replace(".png", ".txt")), "w") as f:
77             f.write(f"{CAMO_CLASS_ID} {bbox[0]:.6f} {bbox[1]:.6f} {bbox[2]:.6f} {bbox[3]:.6f}\n")
78
79         total += 1
80
81     # RGB
82     if os.path.exists(r_path) and os.path.exists(g_path) and os.path.exists(b_path):
83         r = read_gray(r_path); g = read_gray(g_path); b = read_gray(b_path)
84         if r is not None and g is not None and b is not None:
85             rgb = cv2.merge([b, g, r])
86             save_sample(rgb, "_RGB")
87
88     # NIR
89     if os.path.exists(nir_path):
90         nir = read_gray(nir_path)
91         if nir is not None:
92             nir3 = cv2.cvtColor(nir, cv2.COLOR_GRAY2BGR)
93             save_sample(nir3, "_NIR")
94
95     # EIR
96     if os.path.exists(eir_path):
97         eir = read_gray(eir_path)
98         if eir is not None:
99             eir3 = cv2.cvtColor(eir, cv2.COLOR_GRAY2BGR)
100             save_sample(eir3, "_EIR")

```

```
98         if eir is not None:
99             eir3 = cv2.cvtColor(eir, cv2.COLOR_GRAY2BGR)
100             save_sample(eir3, "_EIR")
101
102 print("✅ Total MUDCAD-X samples created:", total)
103
```

```
01_Spring labels found: 328
02_Summer labels found: 254
03_Autumn labels found: 271
✅ Total MUDCAD-X samples created: 2412
```

```
1 yml_text = """
2 path: /content/mudcadx_yolo
3 train: images/train
4 val: images/val
5
6 nc: 1
7 names:
8   0: mudcadx_camouflage_target
9 """
10
11 with open("/content/mudcadx.yaml", "w") as f:
12     f.write(yml_text)
13
14 print("✅ YAML saved at /content/mudcadx.yaml")
15
```

```
✅ YAML saved at /content/mudcadx.yaml
```

```
1 !pip install -q ultralytics
2
```

1.2/1.2 MB 18.7 MB/s eta 0:00:00

```
1 from ultralytics import YOLO
2
```

```

3 model = YOLO("yolo11n.pt") # or yolo11s.pt for better accuracy
4
5 model.train(
6     data="/content/mudcadx.yaml",
7     epochs=5,
8     imgsz=640,
9     batch=8,
10    device='CPU', # use 0 if GPU available, else "cpu"
11    project="/content/runs",
12    name="mudcadx_yolo11"
13 )
14

```

Ultralytics 8.4.11  Python-3.12.12 torch-2.9.0+cpu CPU (AMD EPYC 7B12)

engine/trainer: agnostic_nms=False, amp=True, angle=1.0, augment=False, auto_augment=randaugument, batch=8, bgr=0.0, bo
Overriding model.yaml nc=80 with nc=1

	from	n	params	module	arguments
0	-1	1	464	ultralytics.nn.modules.conv.Conv	[3, 16, 3, 2]
1	-1	1	4672	ultralytics.nn.modules.conv.Conv	[16, 32, 3, 2]
2	-1	1	6640	ultralytics.nn.modules.block.C3k2	[32, 64, 1, False, 0.25]
3	-1	1	36992	ultralytics.nn.modules.conv.Conv	[64, 64, 3, 2]
4	-1	1	26080	ultralytics.nn.modules.block.C3k2	[64, 128, 1, False, 0.25]
5	-1	1	147712	ultralytics.nn.modules.conv.Conv	[128, 128, 3, 2]
6	-1	1	87040	ultralytics.nn.modules.block.C3k2	[128, 128, 1, True]
7	-1	1	295424	ultralytics.nn.modules.conv.Conv	[128, 256, 3, 2]
8	-1	1	346112	ultralytics.nn.modules.block.C3k2	[256, 256, 1, True]
9	-1	1	164608	ultralytics.nn.modules.block.SPPF	[256, 256, 5]
10	-1	1	249728	ultralytics.nn.modules.block.C2PSA	[256, 256, 1]
11	-1	1	0	torch.nn.modules.upsampling.Upsample	[None, 2, 'nearest']
12	[-1, 6]	1	0	ultralytics.nn.modules.conv.Concat	[1]
13	-1	1	111296	ultralytics.nn.modules.block.C3k2	[384, 128, 1, False]
14	-1	1	0	torch.nn.modules.upsampling.Upsample	[None, 2, 'nearest']
15	[-1, 4]	1	0	ultralytics.nn.modules.conv.Concat	[1]
16	-1	1	32096	ultralytics.nn.modules.block.C3k2	[256, 64, 1, False]
17	-1	1	36992	ultralytics.nn.modules.conv.Conv	[64, 64, 3, 2]
18	[-1, 13]	1	0	ultralytics.nn.modules.conv.Concat	[1]
19	-1	1	86720	ultralytics.nn.modules.block.C3k2	[192, 128, 1, False]
20	-1	1	147712	ultralytics.nn.modules.conv.Conv	[128, 128, 3, 2]
21	[-1, 10]	1	0	ultralytics.nn.modules.conv.Concat	[1]

```

22          -1  1   378880  ultralytics.nn.modules.block.C3k2          [384, 256, 1, True]
23      [16, 19, 22]  1   430867  ultralytics.nn.modules.head.Detect      [1, 16, None, [64, 128, 256]]
YOLO11n summary: 182 layers, 2,590,035 parameters, 2,590,019 gradients, 6.4 GFLOPs

```

Transferred 448/499 items from pretrained weights

Freezing layer 'model.23.dfl.conv.weight'

train: Fast image access  (ping: 0.0±0.0 ms, read: 3894.2±1229.9 MB/s, size: 477.2 KB)

train: Scanning /content/mudcadx_yolo/labels/train.cache... 1974 images, 0 backgrounds, 0 corrupt: 100%  1

augmentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3, 7)), ToGray(p=0.01, method='weighte

val: Fast image access  (ping: 0.0±0.0 ms, read: 4464.6±1083.3 MB/s, size: 512.8 KB)

val: Scanning /content/mudcadx_yolo/labels/val.cache... 438 images, 0 backgrounds, 0 corrupt: 100%  438/43

Plotting labels to /content/runs/mudcadx_yolo112/labels.jpg...

optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and 'momentum=0.937' and determining best 'optimizer', 'lr0' an

optimizer: AdamW(lr=0.002, momentum=0.9) with parameter groups 81 weight(decay=0.0), 88 weight(decay=0.0005), 87 bias(

Image sizes 640 train, 640 val

Using 0 dataloader workers

Logging results to /content/runs/mudcadx_yolo112

Starting training for 5 epochs...

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
1/5	0G	1.793	4.765	1.117	12	640: 100%  247/247 3.8s/it 15:49
	Class	Images	Instances	Box(P	R	mAP50 mAP50-95): 100%  28/28 3.1s
	all	438	438	0.436	0.192	0.195 0.133

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
2/5	0G	1.772	3.142	1.124	8	640: 100%  247/247 3.8s/it 15:41
	Class	Images	Instances	Box(P	R	mAP50 mAP50-95): 100%  28/28 3.1s
	all	438	438	0.493	0.393	0.378 0.253

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
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```

1 !cp /content/runs/mudcadx_yolo11/weights/best.pt \
2 /content/drive/MyDrive/major_project/mudcadx_best.pt
3

```

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